

Problem-Solution Fit

This document maps each identified problem to OptiCrop's specific solution feature, demonstrating that the product directly addresses real user pain points.

Problem-Solution Mapping

Problem	Pain Level	OptiCrop Solution	Evidence of Fit
Farmers lack access to crop advisory due to no nearby agronomist	Critical	Web-based prediction accessible on any smartphone	67% of Indian farmers own a smartphone (NSSO 2023)
Soil testing is expensive and geographically inaccessible	High	Accepts estimated NPK from government Soil Health Card	India has issued 220M+ soil health cards (GoI 2023)
Crop selection based on tradition / neighbour's choice	High	ML model trained on 2,200 labelled samples	99.09% accuracy across 22 crops Random Forest test set
No tool combines soil + climate signals for recommendation	High	7-parameter model: N, P, K, temp, humidity, pH, rainfall	All 7 features are globally available / measurable
Farmers distrust complex interfaces	Medium	Single-page form, 7 inputs, 1 button, instant result	80%+ task completion in usability prototype test
Researchers need automated multi-crop dataset analysis	Medium	Jupyter notebooks with EDA, model comparison, metrics	All training notebooks committed to GitHub

Value Proposition Canvas

Customer Jobs

- " Identify the most productive crop for their specific soil and climate conditions
- " Minimise financial risk of crop failure
- " Access agricultural expertise without visiting an agronomist

Customer Pains

- " No affordable soil testing estimate parameters from Soil Health Card
- " No trusted digital tool farmers rely on unreliable word-of-mouth
- " Complex agronomy knowledge simplified to 7-input form

Customer Gains

- " Confident, data-backed crop recommendation in under 2 seconds
- " Increased yield and income from an optimal crop choice
- " Time saved no waiting for expert callback or lab results

OptiCrop Pain Relievers

- " Accepts estimated NPK does not require expensive lab test
- " Mobile-first Bootstrap UI works on budget Android phones
- " Sub-2-second response no waiting for expert callback

OptiCrop Gain Creators

- " Random Forest model achieves 99.09% accuracy on 22-crop dataset
- " Confidence score gives farmer honest uncertainty information
- " Top-3 alternatives shown when primary crop seeds unavailable